CHARAN N N

Full Stack Developer · Fintech & AI Systems · Bengaluru, India
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PROFESSIONAL SUMMARY

Full Stack Developer with 2+ year building and shipping production systems in regulated fintech and AI. Engineered a zero-downtime migration of ~10,000 live cardholders (VISA → RuPay) as the sole engineer; designed a 2-tier automated incident-response pipeline that reduced MTTD to near-zero; cut core API latency by 30%; and built a production AI document-intelligence system using GPT-4 Vision with a 4-agent map-reduce pipeline, guardrails, cost controls, and SHA-256 caching. Works across the full stack — MySQL query optimization, Node.js backend design, SvelteKit frontends, and Docker-based multi-node deployments — and interfaces directly with bank partners, M2P engineering teams, and client stakeholders from BRD through production sign-off. Strong DSA foundations: 1,200+ problems solved across LeetCode (368: 151E/198M/19H) and GeeksforGeeks (844 problems, Coding Score 2500).

PROFESSIONAL EXPERIENCE

Software Engineer
Mar 2025 – Present

Spense Reliable Fintech Solutions · Bengaluru, India

- **VISA-to-RuPay Card Migration (~10,000 live users):** Engineered a zero-downtime migration of ~10,000 VISA cardholders to RuPay without prior tooling or templates. Extracted user profiles absent from our DB by querying Bank APIs using CIF numbers, entity IDs, and FD account numbers supplied by M2P. Built a fault-tolerant multi-step cron pipeline that — on user consent — blocked the VISA card, closed the account, and provisioned a new RuPay card with identical FD details, ensuring migrated users had the same experience as fresh onboardings. Resolved cross-team API inconsistencies during UAT and coordinated final sign-off with bank stakeholders.
- **24/7 Automated Incident-Response System (2-Tier):** Designed a production observability pipeline without prior tooling. Tier 1: a cron cross-referenced mail logs and a failures table to compute live API failure rates; breaches above a configurable threshold (default 10%) auto-dispatched logs to the POC via email, eliminating manual monitoring. Tier 2: critical DB failures triggered Twilio voice calls to a 5-person on-call rotation at any hour. Result: MTTD reduced to near-zero across all critical production events.
- **Full Delivery Lifecycle Ownership (BRD → Production):** Owned every major module at Spense across the full delivery lifecycle — BRD analysis, feasibility, implementation, QA, and UAT. Acted as the sole technical contact during client UAT cycles, resolving integration blockers on-site and coordinating final sign-off with bank stakeholders. Deployed via artifact-based Docker pipeline: built and transferred images to internal servers, orchestrated multi-node rollouts with docker-compose, and validated each release with post-production smoke tests.
- **EMI Conversion Module (M2P Banking Integration):** Built the EMI module for secured credit card transactions, integrating M2P APIs and automating EMI conversion logic across bank, partner, and user. Delivered a seamless user-facing frontend alongside the backend pipeline; adoption tracked live on the partner analytics dashboard as a primary revenue driver.
- **RBAC Admin Panel:** Built a granular role-based access control admin panel in SvelteKit for managing partners, users, and sensitive banking data — access-control schema designed in collaboration with the team to meet bank-grade security requirements.
- **Multi-Tenant Partner Journey Builder — 40% Faster Partner Onboarding:** Replaced hardcoded flows with a configuration-driven engine enabling partners to self-serve user journeys with no engineering involvement required, cutting partner time-to-launch by 40%.
- **Business Intelligence Dashboard:** Built a SvelteKit + Chart.js analytics platform with live data tracking the full funnel (FD deposits → card issuance → EMI conversion) with per-tenant and time-range filters. Eliminated manual reporting for bank partners and internal leadership.
- **Backend Performance:** Cut API response latency by 30% across high-traffic endpoints through structured routing and optimized MySQL query design.

Intern — Full Stack Developer
Aug 2024 – Feb 2025

Nokia Networks · Bengaluru, India

- **Improved network automation dashboard report loading speed by 25%** via API-level caching and error-handling on a React.js + Node.js + MongoDB stack.
- **Worked directly with Nokia network engineers** to map test-execution workflows into structured Chart.js + Tailwind CSS dashboard views adopted by the team.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C++,C#
Frontend: React.js, SvelteKit, HTML5, CSS3, Tailwind CSS, Chart.js
Backend: Node.js, Express.js, Django, Flask
Databases: MySQL (query optimization, indexing), MongoDB
AI & LLM: GPT-4 Vision (OpenAI & Azure), Multi-Agent Orchestration, Prompt Engineering, TensorFlow, OpenCV, LangChain-style pipelines
Systems & Infra: Linux, Cron Jobs, REST API Design, API Security, RBAC, Scalable Architecture
DevOps: Docker, docker-compose, Multi-node Deployment, Artifact-based Release Pipeline
Fintech Domain: M2P Banking APIs, Card Lifecycle Management (VISA/RuPay), Twilio (Voice/SMS), Payment Workflows
Developer Tools: Git, GitHub, Postman, VS Code, GitHub Copilot, pdf-lib
Core CS: Data Structures & Algorithms, System Design, OOP, DBMS

KEY ACHIEVEMENTS

- **~10,000 users migrated** with zero downtime — sole engineer, cross-team coordination across M2P and bank partners.
- **MTTD reduced to near-zero** via a fully automated 2-tier alerting pipeline (cron email + Twilio voice escalation).
- **30% API latency reduction** on production fintech endpoints through structured routing and MySQL optimization.
- **40% faster partner onboarding** via a configuration-driven multi-tenant journey engine.
- **GPT-4 Vision multi-agent pipeline** with guardrails, SHA-256 caching, and HTML sanitization — shipped to production.
- **1,200+ DSA problems** across LeetCode (368: 151E/198M/19H) and GeeksforGeeks (844, Coding Score 2500). LeetCode 100-Day streak badge.

PROJECTS

AI Dynamic Form Intelligence System

- **Built a GPT-4 Vision-powered 4-agent pipeline** (Text / Checkbox / Date-Signature / Validator agents in parallel) that reads any banking PDF and auto-extracts every input field — text, checkbox, radio, table, signature, date — with SHA-256 caching, confidence-based filtering, Levenshtein dedup, and HTML sanitization guardrails. Outputs a live editable web form; user answers written back onto the original PDF at AI-detected coordinates via pdf-lib.

Stack: *Node.js, TypeScript, GPT-4 Vision (OpenAI/Azure), pdf-lib, MongoDB, React.js*

AI-Based Aadhaar Verification System (Major Project)

- **95% face-matching (FaceNet) and 92% voice-matching (MFCC + Cosine Similarity)** — biometric auth system for users unable to use fingerprints/retina scans. Full-stack: React.js + Flask + TensorFlow.

Stack: *Python, Flask, React.js, TensorFlow, OpenCV*

Annamrutha – Food Donation & Management Platform

- **End-to-end Django app** connecting donors, NGOs, and volunteers with live tracking, receipts, and RBAC authentication.

Stack: *Python, Django, MySQL, JavaScript*

EDUCATION

B.E. — Information Science & Engineering *2021 – 2025*
Siddaganga Institute of Technology · Tumakuru, India
CGPA: 9.03 / 10